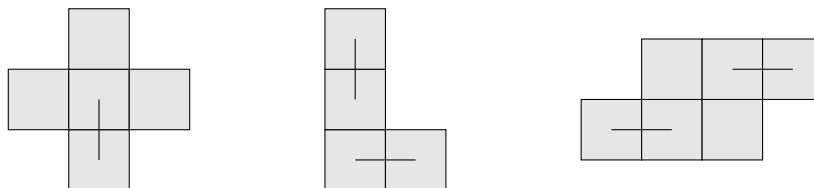


The Fable of the Method of Colouring

by Kevin Haagenen Strömberg

You're doodling in maths class, drawing grids of squares. You start trying to cover them with dominoes and wonder for which grids this is possible to do. You want the grid to be completely covered and for no dominoes to overlap.

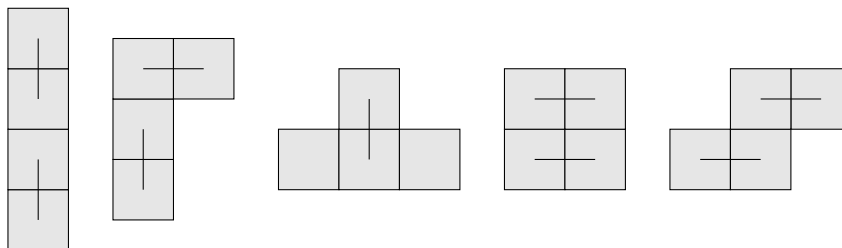


"Clearly there has to be an even number of squares," you think.

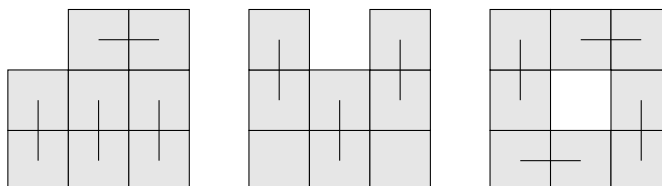
Nothing else seems obvious to you. Some grids work and some don't.

"Let's do some systematic trial and error."

You check all grids with four squares. This doesn't make you wiser.

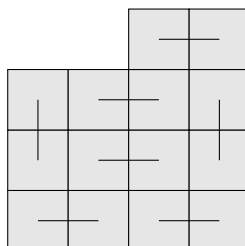


The grids are too different for you to see a pattern, so you try to compare grids with more squares in common. You choose to look at grids obtained by removing one square from a 3×3 grid.



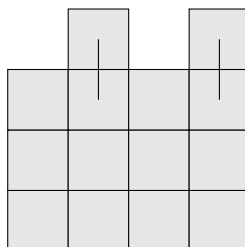
The tileable grids are those obtained by removing either a corner or the centre square. Still, you don't know what to make of this. You choose to look at bigger grids to get more information. You decide to remove pairs of squares from a 4×4 grid. You choose to look at all grids with a removed corner, up to symmetry.

You draw a grid. It turns out that it can be filled with dominoes.

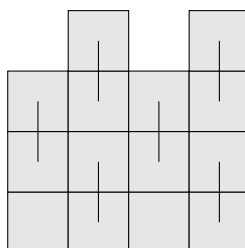


"That was easy."

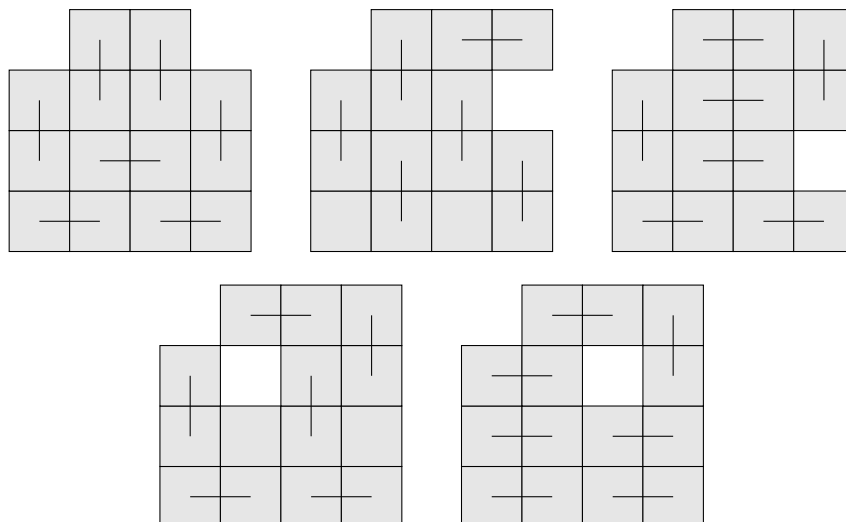
You draw another grid. It contains squares with only a single neighbour. You realise that if the grid could be tiled, there would have to be dominoes there. You cover them.



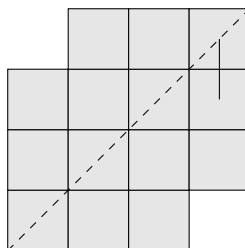
Then you do the same with the remaining squares two more times until the grid is filled.



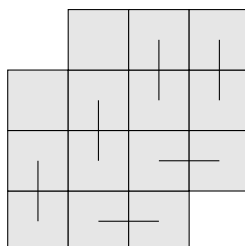
You find five more grids solvable with these methods.



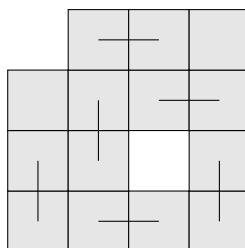
The next grid you look at is trickier. It has no squares with a single neighbour. You realise however that the grid has a symmetry along the NE-SW diagonal. Therefore it doesn't matter how the domino in the top right corner is oriented. You place it vertically.



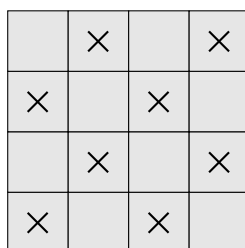
Now the grid has a square with a single neighbour, so you place a domino there. This happens three more times until the grid is clearly impossible to fill.



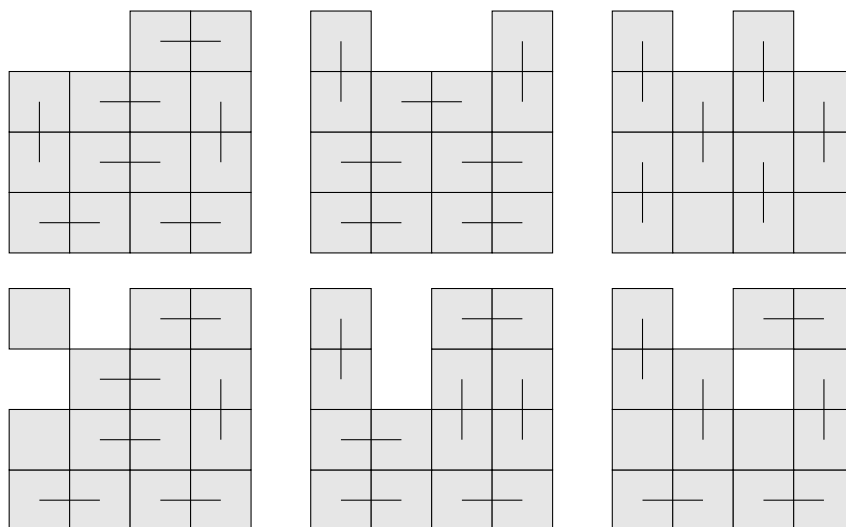
Another grid is also solvable with this symmetry method.

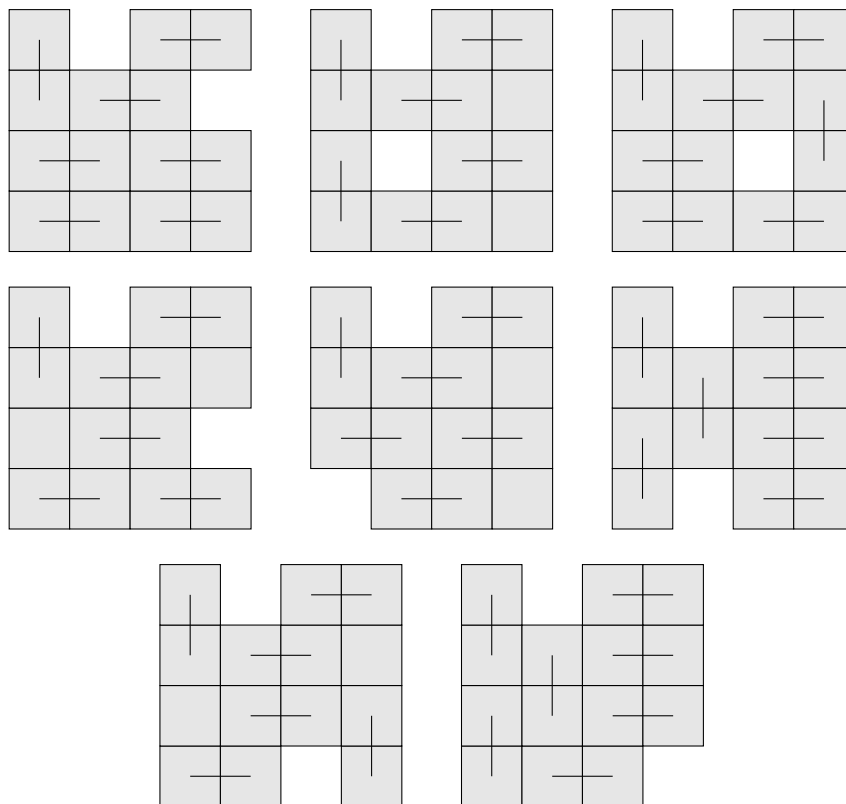


Now that you've looked at all grids with a removed corner, you note down where the second square you removed was for the grid to be tileable. The squares form a checkerboard pattern.



To see if this is just a fluke, you do the same thing with the second square on the first row. The methods you've come up with are sufficient to solve all the grids.

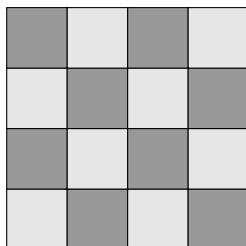




You mark the second square you removed to make each tileable grid. They form the opposite checkerboard pattern. Clearly the checkerboard pattern is important.

×		×	
	×		×
×		×	
	×		×

You colour a checkerboard pattern and stare at it for a minute. Then it hits you.



"Aha! Each domino covers one light and one dark square. Hence the total number of light and dark squares must be equal if the grid is to be tileable."

On a piece of paper you write:

Suppose a standard 8×8 chessboard has two diagonally opposite corners removed. Is it possible to place 31 dominoes so as to cover all of these squares? Solution on the back.

On the back you write:

No! Colour the squares black and white in a checkerboard pattern. There are 30 squares of one colour and 32 of the other. Since each domino covers one square of each colour it is impossible to cover the board.

You pass the note to your friend.